

Tarea corta #4

- Book exercises

12. Match the layers—Link, Network, and Transport—with the guarantees that each layer could provide to higher layers

Garantee	Layer
Best effort delivery	Network
Reliable delivery (confiable)	Transport
In-order delivery	Transport
Byte-stream abstraction	Transport
Point-to-point link abstraction	Data link

13. Suppose that two network endpoints have a round-trip time of 100 milliseconds, and that the sender transmits five packets every round trip. What will be the sender's transmission rate for this round-trip time, assuming 1500-byte packets? Give your answer in bytes per second.

13.

- Round trip time: $100\text{ms} = 0.1\text{s}$
- 5 paquetes por Round trip time.
- Cada paquete de 1500 bytes.

RANGO:

$$(\text{ancho de banda}) = \frac{\text{cant. bytes (RTT)}}{\text{duración (RTT)}} = \frac{5 \times 1500 \text{ bytes}}{0.1\text{s}} = 75000 \text{ bytes/s.}$$

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The subnet of Fig. 1-12(b) was designed to withstand a nuclear war. How many bombs would it take to partition the nodes into two disconnected sets? Assume that any bomb wipes out a node and all of the links connected to it.

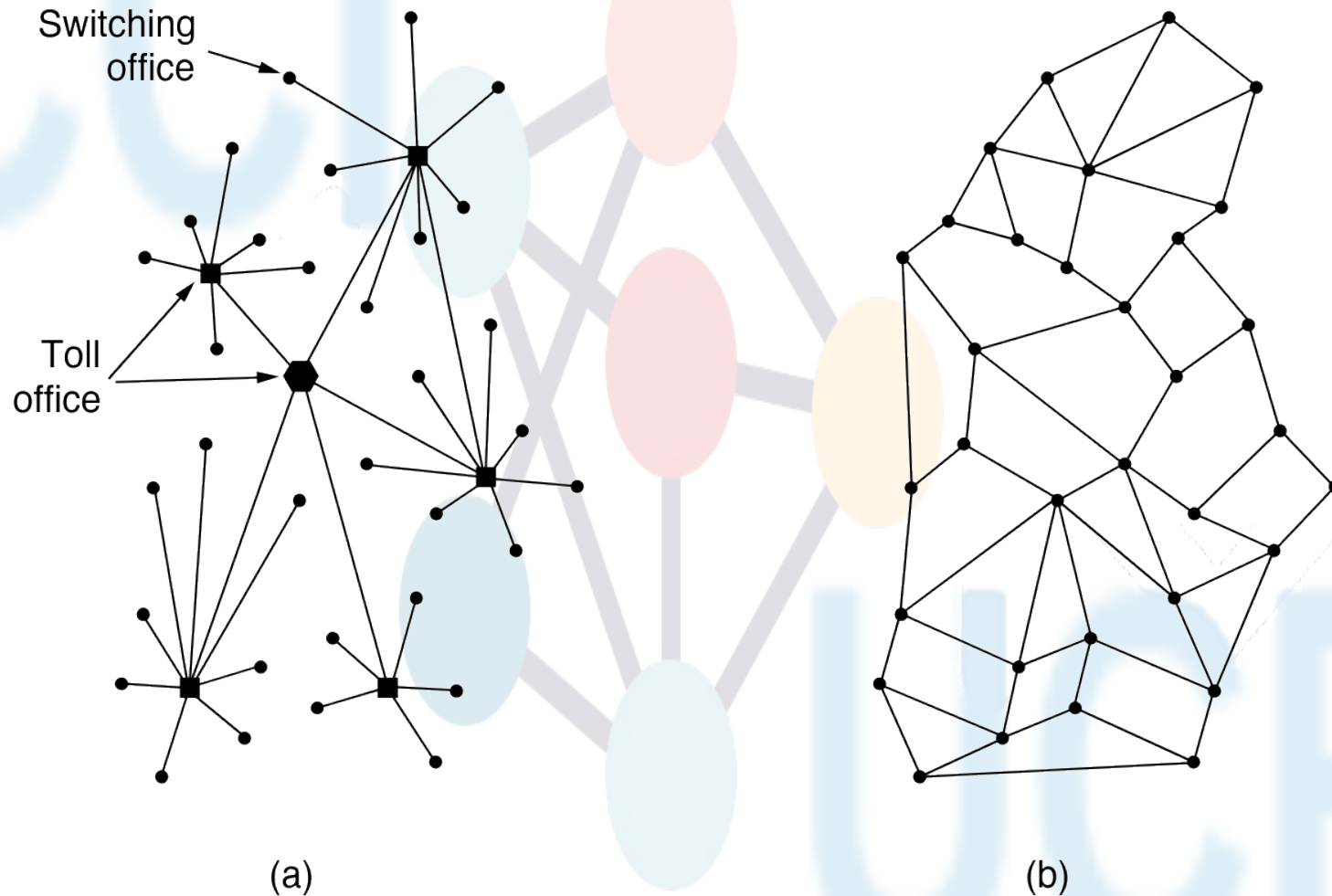


Figure 1-12. (a) Structure of the telephone system. (b) Baran's proposal.

Para separar esta red en dos partes que no se comuniquen entre sí, hay que eliminar todos los nodos de un mismo nivel, como todos las oficinas principales (Toll Offices) o todas las oficinas de cambio (Switching Offices). En este caso hay 2 oficinas principales conectadas entre ellas, aunque eliminemos esas dos oficinas, la red seguiría conectada, por ende no podría dividirse. Como cada bomba destruye un nodo y todos sus enlaces, harían falta 3 bombas para lograr dividir la red en dos partes y del mismo tamaño.